

APPENDIX A

Graduation Project Initial Proposal Template

This template must be completed by the project team and submitted via the faculty platform within the registration period (see Article 5 of the Bylaws). All sections are mandatory unless marked optional.

Section 1 — Project Identification

Project Information

Project Title	<i>Risk-Aware Predictive Maintenance for Aircraft Engines Using NASA C-MAPSS Data and Mixed-Integer Optimization</i>
Department(s)	<i>DS</i>
Academic Year	<i>2026</i>
Semester of Registration	<i>Semester 1</i>
Project Category	<i>AI & optimization</i>

Team Members (Min. 4 and Max. 6)

Student 1	<i>Hager Mahmoud 20221236 DS hm658216@gmail.com</i>
Student 2	<i>Samia reda 20231240 DS samiaareda2004@gmail.com</i>
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Student 4	<i>Mariam Muhammed 20220332 DS mormor.mh2004@gmail.com</i>

Supervision

Primary Supervisor	<i>Dr. Tarek Abu Eleinen Ds</i>
Teaching Assistant (if any)	<i>Eng: Afaf Hamada</i>

Section 2 — Project Description

Motivation and Problem Statement

Aircraft engines are complex systems that experience progressive degradation during operation. Unexpected engine failures can result in flight delays, increased maintenance costs, safety risks, and inefficient resource utilization. Traditional maintenance approaches are typically based on fixed inspection schedules and do not adequately consider the actual health condition of engine components. This project addresses these challenges by utilizing the publicly available NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset, which contains multivariate sensor measurements collected from simulated aircraft engines operating under different conditions and fault modes.

By combining Remaining Useful Life (RUL) prediction with optimization techniques, the proposed framework aims to support intelligent and data-driven maintenance decision-making for aircraft engines

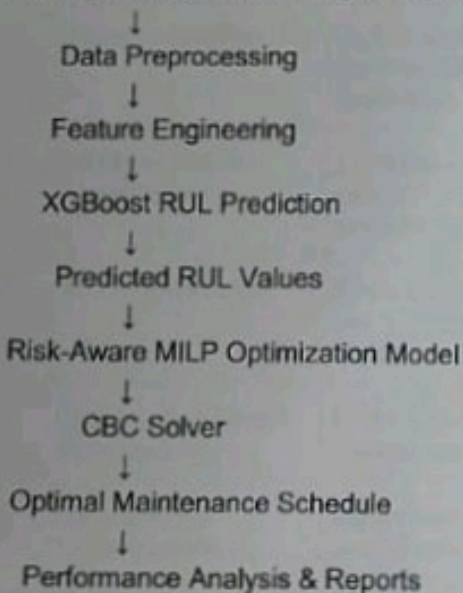
Proposed Solution / Project Objective

The proposed framework consists of two integrated stages. First, sensor measurements from the NASA C-MAPSS aircraft engine dataset are preprocessed and used to train an XGBoost model for predicting the Remaining Useful Life (RUL) of aircraft engines. The predicted RUL values are then employed as inputs to a risk-aware Mixed-Integer Linear Programming (MILP) model.

The optimization model, solved using the CBC solver, generates optimal maintenance schedules and maintenance resource allocation decisions while satisfying operational and risk-related constraints. The framework aims to reduce maintenance costs and unexpected failures, improve aircraft engine reliability and availability, and support predictive maintenance strategies in aviation systems.

Initial High-Level Block Diagram or System Architecture Overview

NASA C-MAPSS Aircraft Engine Dataset



Key Technologies and Tools

NASA C-MAPSS Aircraft Engine Dataset, Python, XGBoost, Pandas, NumPy, Scikit-learn, Mixed-Integer Linear Programming (MILP), Highs Solver, Jupyter Notebook, Microsoft Excel, Matplotlib

Stakeholders and Beneficiaries

Airlines and aviation companies.

Aircraft maintenance organizations and maintenance service providers.

Maintenance engineers and technicians responsible for engine inspection and servicing.

Fleet operation and maintenance managers.

Reliability and safety departments in aviation organizations.

Researchers and academics in predictive maintenance, machine learning, and operations research.

Aircraft manufacturers and engine manufacturers interested in data-driven maintenance strategies

Section 3 — Project Plan

Complete the table below with planned tasks, timeline, and responsibilities.

#	Task Title	Description	Responsible Student(s)
1	Literature Review	Review predictive maintenance, aircraft engine degradation, RUL prediction, and optimization approaches.	Aya-Hager
2	NASA C-MAPSS Data Analysis	Understand and analyze the NASA C-MAPSS aircraft engine dataset and sensor measurements.	All members
3	Data Preprocessing	clean, transform, and prepare the dataset for machine learning modeling.	Mariam
4	Feature Engineering	Select and construct relevant features for accurate RUL prediction.	Mariam -Hager
5	XGBoost Model Development	Develop and train the XGBoost model for Remaining Useful Life prediction.	Aya-samia
6	Model Evaluation	Evaluate prediction performance using appropriate metrics such as RMSE and MAE.	Samia-Aya
7	optimization Model Formulation	Formulate the risk-aware MILP maintenance optimization model.	All members
8	Optimization Implementation	Implement and solve the optimization model using the CBC Solver.	All Members
9	Integration and Testing	Integrate the RUL prediction model with the optimization framework and validate results.	Hager-Mariam
10	Documentation and Final Presentation	Prepare the final report, presentation, and project demonstration.	All members

We, the undersigned, declare that the information provided in this proposal is accurate and that we commit to adhering to the Unified Graduation Project Bylaws of the Faculty of Computers and Artificial Intelligence, Cairo University.

Student Signatures	Supervisor Approval
Student 1: _____	Supervisor Name: _____
Student 2: _____	Signature: _____
Student 3: _____	Date of Approval: _____
Student 4: _____	Department Head Endorsement
Student 5: _____	Signature: _____
	Date: _____